A $U(N)$ Gauge Theory in Three Dimensions as an Ensemble of Surfaces

Source-backed arXiv sample

Abstract. A particular $U(N)$ gauge theory defined on the three dimensional dodecahedral lattice is shown to correspond to a model of oriented self-avoiding surfaces. Using large N reduction it is argued that the model is partially soluble in the planar limit.

Source Metadata

This sample is based on arXiv record `hep-th/9108015`. Authors: F. David, H. Neuberger.

Extracted Prose 1

RU–91–34 July 29, 1991 1.5cm A math expression Gauge Theory in Three Dimensions as an Ensemble of Surfaces 1cm by 1cm Fran cois David Physique Th' eorique CNRS. .2cm Service de Physique Th' eorique Laboratoire de l'Institut de Recherche Fondamentale du Commissariat ` a l'Energie Atomique. de Saclay F-91191 Gif-sur-Yvette Cedex .7cm and .7cm

Extracted Prose 2

1cm A particular math expression gauge theory defined on the three dimensional dodecahedral lattice is shown to correspond to a model of oriented self-avoiding surfaces. Using large math expression reduction it is argued that the model is partially soluble in the planar limit. .3cm